

Machine Learning

Linear regression & Logistic regression
with statsmodel
14/02/2025

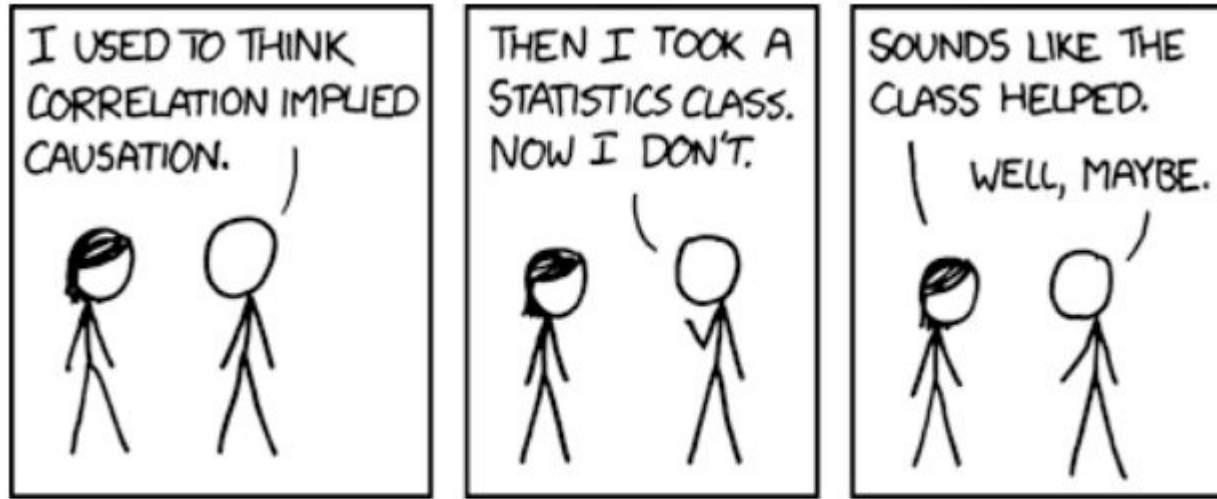
Last time

- What is Machine Learning
- classification vs regression
- supervised vs unsupervised
- data science cycle
- training a model
- scikit-learn
- kaggle
- titanic

Today

- linear regression - statsmodel
 - [univariate](#) and [multivariate](#)
 - interpreting the results
 - p-value, R^2 , t-test, ...
- 5 [conditions](#) for linear regression
- [logistic](#) regression - statsmodel
 - classification metrics
 - dealing with imbalanced datasets
 - multi class classification
- diff between statsmodels and scikit-learn

Correlation and Causation



XKCD correlation and causation <https://www.xkcd.com/552/>